

Date	15 March 2026
Team ID	XXXXXX
Project Name	VBCUA — Voice-Based Concept Understanding Analyser
Maximum Marks	5 Marks

NO. OF FUNCTIONAL FEATURES INCLUDED IN THE SOLUTION

FUNCTIONAL FEATURES OVERVIEW:

This document lists all the functional features that have been included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution. The number and quality of functional features reflect the depth and completeness of the project.

FUNCTIONAL FEATURES BLUEPRINT TABLE

S.No	Feature Name	Feature Description	Module / Component	Status
1	Audio File Upload	Allows learners to seamlessly upload spoken practice explanation files in standard wav, mp3, and m4a formats via web console controls.	app.py (UI Layer)	Done
2	Acoustic Waveform Plot	Renders visual waveform graphs of the uploaded speech stream directly inside the active interface workspace.	app.py / Matplotlib	Done
3	Automated Voice STT	Loads localized speech checkpoints to transcribe spoken sentences into structured text string blocks.	transcription.py	Done
4	Semantic Analysis	Utilizes vector distance checkers to compute contextual compliance rates against selected technical reference cards.	semantic_analysis.py	Done
5	Fluency Diagnostics	Scans acoustic frequency signals to tally silences, filter words, and establish exact pacing speed metrics (WPM).	audio_analysis.py	Done
6	Generative Review Engine	Interacts with external API keys to construct granular concept-gap outlines and targeted practice advice points.	feedback.py	Done
7	Rule-Based Fault Fallback	Ensures complete scoring pipeline continuation by engaging local text heuristics if external network APIs drop out.	feedback.py (Fallback)	Done

S.No	Feature Name	Feature Description	Module / Component	Status
8	Relational History Log	Saves persistent individual test matrices and score grids to a standalone database file layer for profile tracking.	database.py	Done

FEATURE SUMMARY

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

FEATURE CATEGORY BREAKDOWN

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Audio file selection panels, graphical view tabs	Streamlit main layout, Waveform plots
2	Backend / Logic	ML scoring calculations, composite metric equations	Composite weighting scoring logic
3	Database / Storage	Relational schema tracking, profile history stores	SQLite file integration via SQLAlchemy
4	API / Integration	External foundational model keys routing, transcription mappers	Whisper, Sentence-BERT, Gemini API
5	Security / Authentication	Environmental runtime separation policies	Isolated .env file configuration routing